This assignment will not be graded and is only for practice.

Multiple Select Questions (MSQ):

1) Match the systems of linear equations in Column A with their number of solutions in Column B and their geometric representation in Column C. ***1 point***

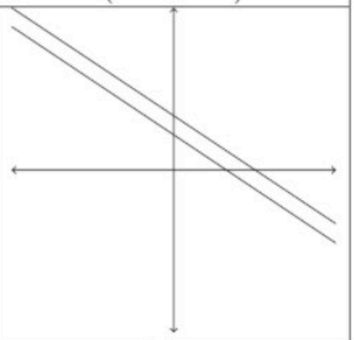
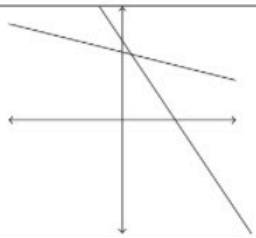
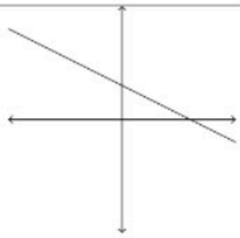
	System of linear equations (Column A)		Number of solutions (Column B)		Geometric representations (Column C)
i)	$3x + 2y = 7,\ x + 4y = 12$	a)	Infinite solutions	1)	
ii)	$2x + 3y = 5,\ 8x + 12y = 13$	b)	No solution	2)	
iii)	$x + 2y = 3,\ 4x + 8y = 12$	c)	Unique solution	3)	

Table: M2W1PT1

- [ ] i) → c) → 2)
- [ ] i) → a) → 1)
- [ ] ii) → c) → 3)
- [ ] iii) → a) → 2)
- [ ] ii) → b) → 1)

No, the answer is incorrect.
Score: 0

Accepted Answers:

i) → c) → 2)
ii) → b) → 1)

2) Choose the set of correct options **1 point**

- [ ] If both A and B are 2×2 real matrices and $det(AB) = 0$, then $det(A) = 0$ or $det(B) = 0$.
- [ ] If A is a 3×3 real matrix with non-zero determinant and k is some real number, then $det(k\,A) = k^3 \times det(A)$.
- [ ] If $A = \begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix}$, then $A^{10} = 2^{10} \begin{bmatrix} 1 & 10 \\ 0 & 1 \end{bmatrix}$ (where A^n is the matrix $A \times A \times \ldots \times A$, n-times).
- [ ] The number of scalar additions to be done to compute the matrix AB, where A is a 3×2 matrix and B is a 2×3 matrix, is 9.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If both A and B are 2×2 real matrices and $det(AB) = 0$, then $det(A) = 0$ or $det(B) = 0$.
If A is a 3×3 real matrix with non-zero determinant and k is some real number, then $det(k\,A) = k^3 \times det(A)$.
If $A = \begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix}$, then $A^{10} = 2^{10} \begin{bmatrix} 1 & 10 \\ 0 & 1 \end{bmatrix}$ (where A^n is the matrix $A \times A \times \ldots \times A$, n-times).
The number of scalar additions to be done to compute the matrix AB, where A is a 3×2 matrix and B is a 2×3 matrix, is 9.

3) Choose the set of correct options **1 point**

- [ ] A triangular 3×3 matrix has non-zero determinant if and only if all the diagonal entries are non-zero.
- [ ] If A and B are 3×3 matrices then $det(A + B) = det(A) + det(B)$.
- [ ] If A is a 3×3 matrix and B is a matrix obtained from A by multiplying each column of A by its column number, then $det(B) = 6det(A)$.

- [ ] If the sum of the first and the third row vectors of a 3×3 matrix A is equal to the second row vector of A, then $det(A) = 0$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

A triangular 3×3 matrix has non-zero determinant if and only if all the diagonal entries are non-zero.
If A is a 3×3 matrix and B is a matrix obtained from A by multiplying each column of A by its column number, then $det(B) = 6det(A)$.
If the sum of the first and the third row vectors of a 3×3 matrix A is equal to the second row vector of A, then $det(A) = 0$.

4) Mahesh bought 2 kg potato and c kg dal from a shop, and paid ₹ 200 to the shopkeeper. Gaurav bought 4 kg potato and 4 kg dal, and paid ₹ d to the shopkeeper. If x_1 represents the price of 1 kg potato and x_2 represents the price of 1 kg dal, then choose the set of correct options. ***1 point***

- [ ] The matrix representation to find x_1 and x_2 is $\begin{bmatrix} 2 & 4 \\ 4 & c \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 200 \\ d \end{bmatrix}$
- [ ] The matrix representation to find x_1 and x_2 is $\begin{bmatrix} 2 & c \\ 4 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 200 \\ d \end{bmatrix}$
- [ ] If Mahesh tries to find the price of 1 kg potato and 1 kg dal using appropriate matrix representation by taking $c = 2$ and $d = 400$, then the price of 1 kg potato that he thus arrives at, will not be unique.
- [ ] If Mahesh tries to find the price of 1 kg potato and 1 kg dal using appropriate matrix representation by taking $c = 2$ and $d = 400$, then the price of 1 kg potato that he thus arrives at, will be unique.
- [ ] If Mahesh tries to find the price of 1 kg potato and 1 kg dal using the appropriate matrix representation by taking $c = 2$ and $d = 400$, then he will be able to find the price (as a numerical value) of 1 kg potato.
- [ ] If Mahesh tries to find the price of 1 kg potato and 1 kg dal using the appropriate matrix representation by taking $c = 2$ and $d = 400$, then he will fail to find the price (as a numerical value) of 1 kg potato.

No, the answer is incorrect.
Score: 0

Accepted Answers:

The matrix representation to find x_1 and x_2 is $\begin{bmatrix} 2 & c \\ 4 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 200 \\ d \end{bmatrix}$

If Mahesh tries to find the price of 1 kg potato and 1 kg dal using appropriate matrix representation by taking $c = 2$ and $d = 400$, then the price of 1 kg potato that he thus arrives at, will not be unique.

If Mahesh tries to find the price of 1 kg potato and 1 kg dal using the appropriate matrix representation by taking $c = 2$ and $d = 400$, then he will fail to find the price (as a numerical value) of 1 kg potato.

5) The marks obtained by Safina, Ram and Pratiksha in Quiz 1, Quiz 2 and End sem (with the maximum marks for each exam being 100) are shown in Table W1PT2. ***1 point***

	Quiz 1	Quiz 2	End sem
Safina	89	95	88
Ram	92	81	98
Pratiksha	85	93	98

Table: W1PT2

The weightage of marks in final grade(in percent) of Quiz 1, Quiz 2, and End sem is shown in Table W1PT3.

	In percent (%)
Quiz 1	20
Quiz 2	20
End sem	60

Table: W1PT3

Choose the set of correct options.

- [ ] Final grades (in 100) of Safina, Ram and Pratiksha can be represented by the matrix:

$$\begin{bmatrix} 93.4 \\ 94.4 \\ 89.6 \end{bmatrix}$$

- [ ] Final grades (in 100) of Safina, Ram and Pratiksha can be represented by the matrix :

$$\begin{bmatrix} 89.6 \\ 93.4 \\ 94.4 \end{bmatrix}$$

- [ ] The order of the matrix which represents final grades(in 100) of Safina, Ram and Pratiksha is 3×1

- [ ] If bonus marks given to Safina, Ram and Pratiksha are represented by the following matrix $\begin{bmatrix} 1.4 \\ 2.6 \\ 0 \end{bmatrix}$ then the overall final grades (in 100) can be represented by the matrix:

$$\begin{bmatrix} 91 \\ 96 \\ 94.4 \end{bmatrix}$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

Final grades (in 100) of Safina, Ram and Pratiksha can be represented by the matrix :

$$\begin{bmatrix} 89.6 \\ 93.4 \\ 94.4 \end{bmatrix}$$

The order of the matrix which represents final grades(in 100) of Safina, Ram and Pratiksha is 3×1

If bonus marks given to Safina, Ram and Pratiksha are represented by the following matrix $\begin{bmatrix} 1.4 \\ 2.6 \\ 0 \end{bmatrix}$ then the overall final grades (in 100) can be represented by the matrix:

$$\begin{bmatrix} 91 \\ 96 \\ 94.4 \end{bmatrix}$$

Numerical Answer Type (NAT):

6) Let A be a 3×3 matrix with non-zero determinant and B be a matrix obtained by adding 5 times of first row of$A$ to the third row of A and adding 10 times of second row of A to the first row of A.What is the value of $det(2AB^{-1})$?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 8

1 point

7) If $A = \begin{bmatrix} 4 & 6 & 10 \\ 1 & 4 & -5 \\ 4 & 1 & 10 \end{bmatrix}$,then find $det(A)$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) -150

1 point

Comprehension Type Question:

(Use the below information for question 8, 9 and 10)

A shopkeeper sells three types of clothes- shirts, jeans, and T- shirts- in three different sizes: small, medium, and large. In a week, he sold 1 small, 1 medium and 2 large sized shirts; 2 small, c medium and 6 large sized jeans, and 1 small, 3 medium and $c - 5$ large sized T-shirts (where c is an integer). The price of shirts, jeans, and T-shirts remain the same for different sizes (i.e., small, medium, and large sized shirts have the same price; similarly, small, medium, large sized jeans have the same price; and small, medium, large sized T-shirts have same price). The shopkeeper earned ₹7 , ₹27 and ₹43 (in thousand) in that week, for small, medium, and large sized clothes respectively.

8) Let s, j, t represents the price (in thousand) of 1 shirt, 1 jeans and 1 T-shirt respectively and we want to find s, j, t by solving a system of linear equations represented by the matrix form $Ax = b$, where $x = (s, j, t)^T$ and $b = (7, 27, 43)^T$. If sum of entries in the first row of A is p, sum of entries in the second row of A is q and sum of entries in the third row of A is r (take the topmost row as the fist row), then find the value of $p + q - r$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 5

1 point

9) If $det(A) = 122$, then how many medium sized jeans were sold in the week?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 16

1 point

10) If A is the matrix as above (in question 8) and $det(A) = 122$, then what is the price(in thousand) of a T-shirt, where price (in thousand) of a shirt is 2 and price (in thousand) of a jeans is 1?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

1 point

Your score is: 0/10